



Q1. An electron, an alpha-particle and a proton have the same kinetic energy. Which one of these particles has the largest de-Broglie wavelength?

Answer: We know that de-Broglie wavelength $\lambda = \frac{h}{mv}$ and Kinetic Energy $= K = \frac{1}{2}mv^2$

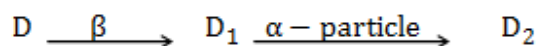
$$\text{or } 2mK = m^2v^2 \text{ or } \sqrt{2mK} = mv, \lambda = \frac{h}{\sqrt{2mK}} \text{ i.e. } \lambda \propto \frac{1}{\sqrt{m}}$$

Since mass of alpha > mass of proton > mass of Electron. Electro possess lowest mass therefore de-Broglie wave length from electron has the largest de-Broglie wavelength.

Q2. Why should the material used for making permanent magnets have high coercivity?

Answer: High coercivity means it can withstand high external magnetic and electric field. Therefore high coercivity means magnetization is not destroyed by stray magnetic fields or temperature variations. For making permanent magnet this property is required. So Ferromagnetic materials with high coercivity used for permanent magnet.

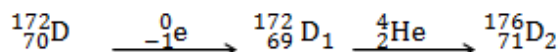
Q3. The radioactive isotope D decays according to the sequence



If the mass number and atomic number of D_2 are 176 and 71 respectively, what is (I) the mass number (II) atomic number of D?

Answer: We know that α emission reduces mass by 4 unit and atomic number by 2 unit, β emission only increases atomic number by 1 unit but brings no change in mass. Hence we can write

$$176 - 4 = X, X = 172, 71 + 1 - 2 = Y, Y = 70$$



(I) Mass number of D = 172

(II) Atomic number of D = 70

Q4. What will be the values of input A and B for the Boolean expression $(\overline{A+B})(\overline{A.B}) = 1$?

Ans. If A=1, B=1 then 0.0=0, if A=1, B=0 then 0.1=0, similarly A=0, B=1 we get 0.0=0 so not possible

Only If A = 0 and B = 0. $(\overline{0+0})(\overline{0.0}) = (\overline{0}).(\overline{0}) = 1.1 = 1$. Answer is A=B=0



Q5. Why is frequency modulation preferred over amplitude modulation for transmission of music?

Ans. In transmission of music we want to avoid noise, noise can be minimized in frequency modulation hence frequency modulation preferred over amplitude modulation.

Q6. Define electromotive force of a cell.

Answer: The potential difference between two poles of a cell, when no current is drawn from it, is called *electromotive force (emf)* of the cell.

Q7. An electron is projected with a velocity of 10^5 ms^{-1} at right angles to a magnetic field of 0.019 G. Calculate the radius of the circular path described by the electron,

if $e = 1.6 \times 10^{-19} \text{ C}$, $m = 9.1 \times 10^{-31} \text{ kg}$.

Answer: Given $v = 10^5 \text{ ms}^{-1}$, $e = 1.6 \times 10^{-19} \text{ C}$, $m = 9.1 \times 10^{-31} \text{ kg}$;

$B = 0.019 \text{ G} = 0.019 \times 10^{-4} \text{ T}$

We know that centrepetal force is obtained from the force due to magnetic field on moving charge

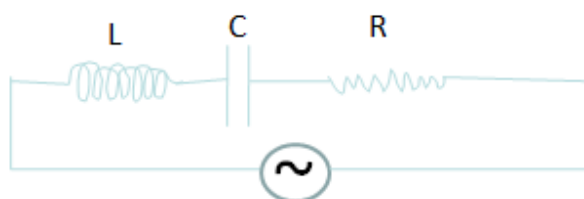
$$\text{Now } Bev = \frac{mv^2}{r}$$

$$\Rightarrow r = \frac{mv}{Be} = \frac{9.1 \times 10^{-31} \times 10^5}{0.019 \times 10^{-4} \times 1.6 \times 10^{-19}} = 0.299 \text{ m}$$

Q8. What do you mean by the impedance of LCR circuit?

Answer: The effective resistance offered to the flow of current by the LCR circuit is called its **impedance**.

L-C-R circuit :



The impedance of the circuit

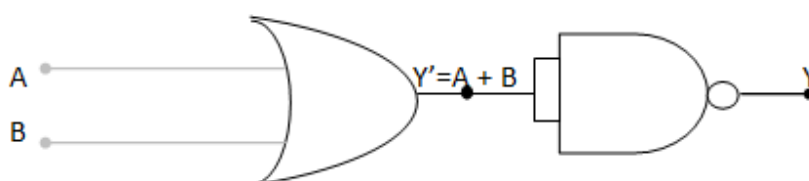
$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

Impedance is denoted by $Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$



Q9. The output of an OR gate is connected to both the inputs of a NAND gate. Draw the logic circuit of this combination of gates and write its truth table.

Answer: Logic Circuit , here output of OR gate is input for the NAND gate

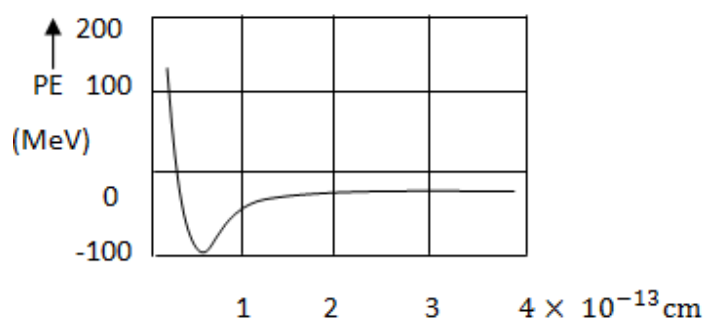


Truth Table for the above Logic Circuit

OR Gate			Input of NAND Gate $Y' = C = D$		AND Operation	NAND Operation
A	B	$Y' = A + B$	C	D	$C.D$	$Y = (\overline{C.D})$
0	0	0	0	0	0	1
0	1	1	1	1	1	0
1	0	1	1	1	1	0
1	1	1	1	1	1	0

Q10. Draw a plot of potential energy of a pair of nucleons as a function of their separation. What is the significance of negative potential energy in the graph drawn?

Answer: Potential energy (MeV) taken along Y axis and distance of separation of pair of nucleons taken along X-axis, graphical presentation is as shown



The negative potential energy indicates the force between the pair of nucleons is attractive.

Q11. A convex lens of refractive index 1.5 has a focal length of 8 cm in air. Calculate the change in its focal length when it is immersed in water of refractive index $4/3$.

Answer: Let f_a and f_w be the focal length in air and water respectively.

$$\begin{aligned} \text{Then } \frac{1}{f_a} &= (\mu_g - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right] \\ \text{or, } \frac{1}{8} &= (1.5 - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right] \\ \text{or, } \left[\frac{1}{R_1} - \frac{1}{R_2} \right] &= \frac{1}{8 \times 0.5} = \frac{1}{4} \end{aligned}$$

When the lens is immersed in water,

$$\begin{aligned} \frac{1}{f_w} &= \left(\frac{\mu_g}{\mu_w} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right] = \left(\frac{1.5}{4/3} - 1 \right) \times \frac{1}{4} \\ \frac{1}{8} \times \frac{1}{4} &= \frac{1}{32} \Rightarrow \frac{1}{f_w} = \frac{1}{32} \quad \therefore f_w = 32 \text{ cm.} \end{aligned}$$

Q12. Distinguish between the terms 'average value' and 'rms value' of an alternating current. The instantaneous current from an a.c. source is $i = 5 \sin (314 t)$ ampere. What are the average and rms values of the current?

Answer: **Average value of AC:** The arithmetic mean of all the instantaneous values of current or emf over the half cycle is known as average value of A.C.



Let $i = i_p \sin \omega t \rightarrow (1)$

The instantaneous value of current at an instant of time t

$$i_{\text{ave}} = \frac{1}{T} \int_0^T i dt$$

Putting equation (1): $i_{\text{ave}} = \frac{1}{T} \int_0^T i_p \sin \omega t dt$

$$i_{\text{ave}} = \frac{2i_p}{T} \left[\frac{-\cos \omega t}{\omega} \right]_0^T$$

$$i_{\text{ave}} = -\frac{2}{T} \frac{i_p}{2\pi} \left[\cos \frac{2\pi T}{2} - \cos 0 \right]$$

$$i_{\text{ave}} = -\frac{i_p}{\pi} [\cos \pi - \cos 0]$$

$$i_{\text{ave}} = \frac{2i_p}{\pi} \rightarrow (2)$$

Similarly

$$e_{\text{ave}} = \frac{2e_p}{\pi} \rightarrow (3)$$

RMS Value of AC: To get a suitable mean value we square all the instantaneous values for a complete cycle where by all become positive and then are added and the mean is taken. Finally to get rid of square the square root is taken. This value is known as root mean square value.

Let $e = e_p \sin \omega t \rightarrow (1)$

The instantaneous emf i.e. the value of emf at an instant of time t

$$e_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T e^2 dt}$$

$$e_{\text{rms}}^2 = \frac{1}{T} \int_0^T e^2 dt$$

$$e_{\text{rms}}^2 = \frac{1}{T} \int_0^T e_p^2 \sin^2 \omega t dt$$

$$e_{\text{rms}}^2 = \frac{e_p^2}{2T} \int_0^T 2 \sin^2 \omega t dt$$

$$e_{\text{rms}}^2 = \frac{e_p^2}{2T} \int_0^T (1 - \cos 2\omega t) dt$$

$$e_{\text{rms}}^2 = \frac{e_p^2}{2T} \left[\int_0^T dt - \int_0^T (\cos 2\omega t) dt \right]$$

$$e_{\text{rms}}^2 = \frac{e_p^2}{2T} \left[T - \frac{1}{2\omega} (\sin 2\omega T - \sin 2\omega \cdot 0) \right]$$



Given $I = 5 \sin (314 t)$ ampere

$\therefore$ Peak value of current, $I_p = 5A$

$$I_{av} = 0.637 I_p = 0.637 \times 5 = 3.185 A$$

$$I_{rms} = 0.707 I_p = 0.707 \times 5 = 3.535 A$$

13. Write the relation for the force $\vec{F}$ acting on a charge carrier q moving with a velocity $\vec{v}$ through a magnetic field $\vec{B}$ in vector notation. Using this relation, deduce the conditions under which this force will be (I) maximum (II) minimum.

Answer: $\vec{F} = q(\vec{v} \times \vec{B})$

$$\therefore F = qvB \sin \theta$$

- (I) When $\theta = 90^\circ$, $F = qvB \sin 90^\circ = qvB$ i.e., force is maximum. Thus when velocity is at right angle to the direction of magnetic field then force is maximum.
- (II) When $\theta = 0^\circ$, $F = qvB \sin 0^\circ = 0$ i.e., force is minimum. Force will be minimum also when $v = 0$. Thus when the velocity is in parallel to the direction of magnetic field no force acts on the charge carrier i.e. force is minimum.



Q14. A cylindrical metallic wire is stretched to increase its length by 5%. Calculate the percentage change in its resistance.

Answer: Considering l = Original length, Changed length is $l' = l + 5\%$ of $l = 1.05l$

Since volume remains same therefore initial volume = final volume

$$Al = A' l'$$

$$\therefore \frac{A}{A'} = \frac{l'}{l} = 1.05, \text{ Since } \therefore R = \rho \frac{l}{A}$$

$$\frac{R'}{R} = \frac{l'}{l} \times \frac{A}{A'} = \left(\frac{l'}{l}\right)^2 = (1.05)^2 = 1.1025$$

Percentage increase in resistance is

$$\frac{R' - R}{R} \times 100 = \left(\frac{R'}{R} - 1\right) \times 100 = (1.1025 - 1) \times 100 = 10.25\%$$

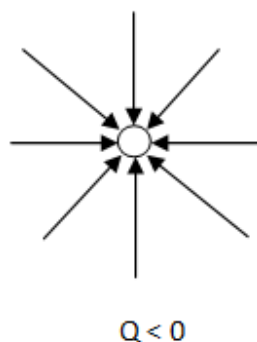
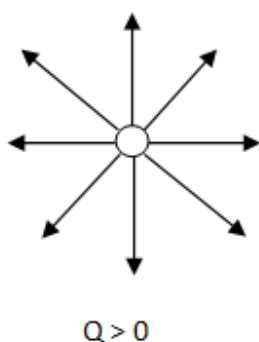
Q15. The electric field E due to a point charge at any point near it is defined as $E = \lim_{q \rightarrow 0} \frac{F}{q}$, where q is the test charge and F is the force acting on it. What is the physical significance of $\lim_{q \rightarrow 0}$ in this expression? Draw the electric field lines of a point charge Q when (I) $Q > 0$ and (II) $Q < 0$.

Answer: The $\lim_{q \rightarrow 0}$ indicates that the test charge (point charge) q is very small hence presence of this point charge does not disturb the distribution of source charge and its electric field.

The electric fields of the point charge Q are shown in the figure.

Positive Charge lines of force coming out

Negative Charge lines of force coming in





Or,

Define electric flux. Write its S.I. units. A spherical rubber balloon carries a charge that is uniformly distributed over its surface. As the balloon is blown up and increases in size, how does the total electric flux coming out of the surface change? Give reason.

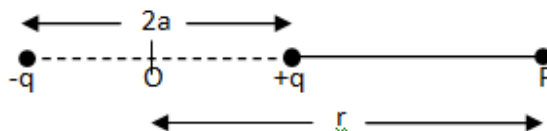
Answer: The electric flux through a given surface area is the total number of electric lines of force passing normally through that area. It is given by $\phi_E = \vec{E} \cdot \vec{\Delta S}$

SI unit of electric flux is $\text{Nm}^2 \text{C}^{-1}$.

As the balloon is blown up, the total charge on the balloon surface remains unchanged, so the total electric flux coming out of its surface remains unchanged.

Q16. Deduce an expression for the electric potential due to an electric dipole at any point on its axis. Mention one contrasting feature of electric potential of a dipole at a point as compared to that due to a single charge.

Answer: Let P be an axial point at distance r from the center of the dipole.



Electric potential at point P will be, For +q charge potential is positive as work is done to bring unit positive charge is against the repulsive force also potential for -q charge is negative as work done in favour of attractive force.

$$\begin{aligned}
 V &= V_1 + V_2 = -\frac{1}{4\pi\epsilon_0} \frac{q}{r+a} + \frac{1}{4\pi\epsilon_0} \frac{q}{r-a} \\
 &= \frac{1}{4\pi\epsilon_0} \left[\frac{1}{r-a} - \frac{1}{r+a} \right] = \frac{q}{4\pi\epsilon_0} \cdot \frac{2a}{r^2-a^2} \\
 &= \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^2-a^2} \quad [\because p = \text{dipole moment} = (q) 2a]
 \end{aligned}$$

For a far away point, $r \gg a$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{p}{r^2}$$

At large distances, dipole potential fall off as $1/r^2$, whereas the potential due to a single charge falls off as $1/r$.



Q17. A parallel plate capacitor, each with plate area A and separation d , is charged to a potential difference V . the battery used to charge it is then disconnected. A dielectric slab of thickness d and dielectric constant K is now placed between the plates. What change, if any, will take place in

- (I) charge on the plates
- (II) electric field intensity between the plates
- (III) capacitance of the capacitor

Justify your answer in each case.

Answer:

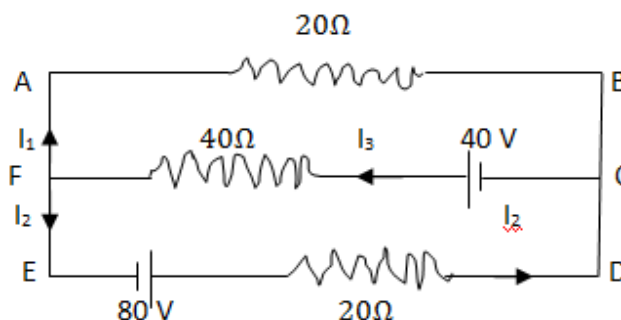
- (I) Introduction of dielectric slab will have no effect on charge on plates of capacitor, hence the charge q on the capacitor plates remains same.
- (II) Because of dielectric slab electric field intensity between the capacitor plates decreases as surface charges induced on the dielectric. Actual field and induced field directions are different hence net electric field decreases.

$$E = \frac{E_0}{K}$$

- (III) $Q=CV$, since V i.e. potential difference decreases, net charge remains constant hence Capacitance C will increase

$$C = K \cdot C_0$$

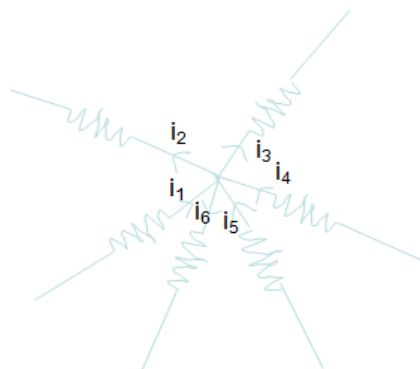
Q18. State Kirchhoff's rules of current distribution in an electrical network. Using these rules determine the value of the current I_1 in the electric circuit given as



Answer: **Kirchhoff's law** : In a complicated networks to find current flowing through different branches we apply Kirchhoff's law.

First law: The algebraic sum of currents meeting at a point is zero. i.e. $\sum i = 0$

Explanation: Let us follow a convention current flowing towards the point is taken as positive and flowing away from the point is taken as negative.



$$\sum i = i_1 - i_2 - i_3 + i_4 + i_5 - i_6 = 0$$

Kirchhoff's first law is also known as point theorem or junction theorem.

Second law : "In a closed mesh of electrical conductors the algebraic sum of product of resistance and the respective current in the different branches is equal to the total e.m.f applied in the closed mesh"

Kirchhoff's second law is also known as loop theorem.

Solution of Numerical: By Kirchhoff's rule, $I_1 + I_2 = I_3$.

Applying loop rule to both the lower and upper loops, we get

$$40 I_3 + 20 I_1 = 40 \quad (\text{In loop ABCF})$$

$$40 I_3 + 20 I_2 = 80 + 40 \quad (\text{In loop CDEF})$$

Adding the two equations, we get

$$80 I_3 + 20 (I_1 + I_2) = 160$$

$$\text{Or, } 80 I_3 + 20 I_3 = 160$$

$$\text{or, } I_3 = \frac{160}{100} = 1.6 \text{ A}$$

$$\text{Again, } 40 \times 1.6 + 20 I_1 = 40$$

$$\text{or, } 20 I_1 = 40 - 64 = -24$$

$$\text{or, } I_1 = -\frac{24}{20} = -1.2 \text{ A}$$

Q19. Write the mathematical relation for the resistivity of a material in terms of relaxation time, number density and mass and charge of charge carriers in it. Explain, using this relation, why the resistivity of a metal increases and that of a semi-conductor decreases with rise in temperature.

Answer: Resistivity, $\rho = \frac{m}{ne^2 \tau}$

- (I) **In Metallic conduction:** As the temperature increases, the thermal speed of electrons increases. Free electrons collide more frequently with the positive metal ions. The relaxation time τ decreases. Consequently, the resistivity ρ of the metal increases.
- (II) **In Semiconductors:** The relaxation time τ does not change with temperature. But the number density (n) of free electrons increases exponentially with temperature. As a result, the resistivity of semiconductor decreases exponentially with the increase in temperature.



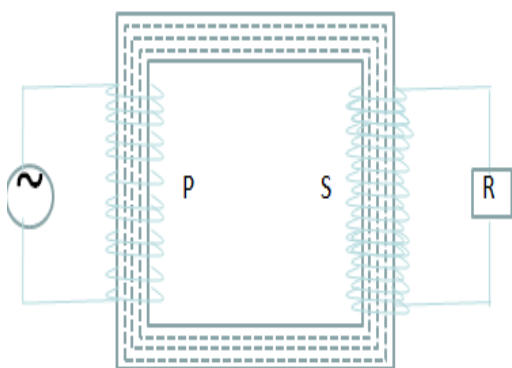
Q20. Explain with the help of a labeled diagram the underlying principle and working of a step up transformer. Why cannot such a device be used to step-up d.c. voltage?

Answer: Transformer is a device for converting high voltage at low current to high current at low voltage or vice versa. The A.C transformer is based on the principle of electromagnetic induction and are basically mutual inductance.

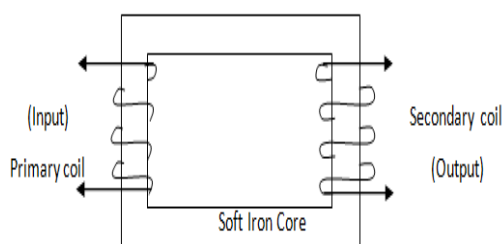
A transformer which increases the A.C. voltage is called a **step up transformer**. A transformer which decreases the a.c. voltage is called **step down transformer**.

Principle: A transformer is based on the principle of mutual induction. (i.e. whenever the amount of magnetic flux linked with a coil changes, an emf is induced in the neighboring coil).

Construction:



It consists of two coils known as Primary (P) and Secondary(S) having different no. of turns and are wound on two opposite arms of a thick laminated iron core. The alternating emf to be transformed is connected across the primary coil. The varying current flowing through the primary coil produces varying magnetic lines of force. The magnetic iron core provides an easy path for the flow of lines of force hence the lines of force flowing through the core cut the secondary coil and induce an emf across the secondary. An induced current flows through the secondary coil and thus a potential drop is obtained across the resistance which is known as output voltage.



Working: When an alternating voltage is applied to the primary coil, the resulting current produces an alternating magnetic flux which links the secondary coil and induces an emf in it. The value of emf depends on the number of turns in the secondary coil.



We consider an ideal transformer. Let ϕ be the flux in each turn in the core at time t due to current in primary coil when voltage V_p is applied. Then induced emf E_s in the secondary coil is

$$E_s = -N_s \frac{d\phi}{dt}$$

The alternating flux ϕ also induced back emf in the primary coil.

$$\therefore E_p = -N_p \frac{d\phi}{dt}$$

But $E_p = V_p$, Since primary coil has zero resistance and $E_s = V_s$, if the current taken in secondary coil is small where V_s is the voltage across secondary coil.

$$\therefore V_s = -N_s \frac{d\phi}{dt} \quad \dots \dots \dots (1)$$

$$\text{and } V_p = -N_p \frac{d\phi}{dt} \quad \dots \dots \dots (2)$$

Dividing (1) by (2), we get

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} \quad \dots \dots \dots (3)$$

If the transformer is assumed to be 100% efficient, then power input is equal to power output.

$$\therefore I_p V_p = I_s V_s$$

$$\text{or, } \frac{I_p}{I_s} = \frac{V_s}{V_p} \quad \dots \dots \dots (4)$$

From (3) and (4), we have

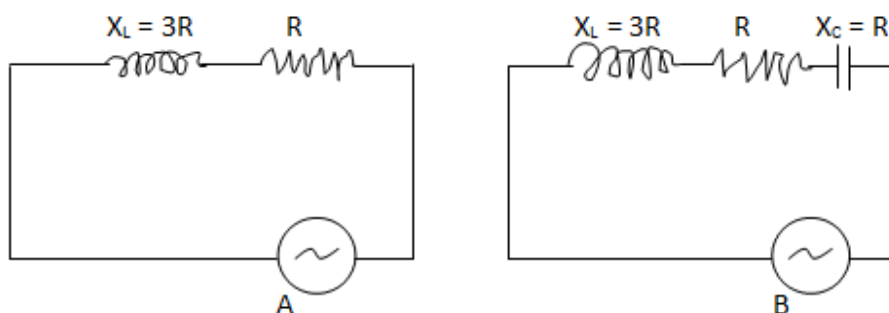
$$\frac{I_p}{I_s} = \frac{V_s}{V_p} = \frac{N_s}{N_p}$$

(II) Why cannot such a device be used to step-up d.c. voltage?

For electromagnetic induction one of the primary condition is magnetic lines of force (flux) should change, if direct current passed then magnetic lines will not change hence mutual induction will not happen, transformer will not work. Thus, the transformer cannot be used to step-up D.C. voltage.



Q21. Given below are two electric circuits A and B



Calculate the ratio of power factor of the circuit B to the power factor of circuit A.

Answer: In LR Circuit A, Impedance $z = \sqrt{R^2 + X_L^2} = \sqrt{R^2 + (\omega L)^2}$ ohm

$$\text{Power factor of A} = \cos \phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + X_L^2}} = \frac{R}{\sqrt{R^2 + 9R^2}} = \frac{R}{\sqrt{10}R} = \frac{1}{\sqrt{10}}$$

In LCR Circuit B, Impedance $Z = z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$

$$\text{Power factor of B} = \cos \phi = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{R}{\sqrt{R^2 + (3R - R)^2}} = \frac{1}{\sqrt{5}}$$

$$\frac{\text{Power factor of B}}{\text{Power factor of A}} = \frac{1}{\sqrt{5}} \times \frac{\sqrt{10}}{1} = \sqrt{2}$$

Q22. Define the term 'resolving power' of an astronomical telescope. How does it get affected on

- (I) Increasing the aperture of the objective lens?
 - (II) Increasing the wavelength of the light used?
- Justify your answer in each case.

Answer: Ability to make us see really small details and see sharp images is resolving power. *The resolving power of an astronomical telescope* is defined as the reciprocal of the smallest angular separation between two point objects whose images can just be resolved by the telescope.

Resolving power depends on Diameter (D) and Wavelength(λ)

$$\text{Resolving Power} = \frac{D}{1.22\lambda}$$

- (I) Since resolving power is proportional to Diameter($\propto D$), when the aperture of the objective lens is increased, the resolving power also increases as diameter in cases.
- (II) Since resolving power is inversely proportional to wavelength ($\propto \frac{1}{\lambda}$), hence if the wavelength of the light is increased, the resolving power decreases.



Q23. Write any four characteristics of electromagnetic waves. Give two uses each of (I) Radio-waves (II) Micro-waves.

Answer: *Characteristics of electromagnetic waves:*

1. The oscillations of electric ($\vec{E}$) and magnetic ($\vec{B}$) field vectors are perpendicular to each other and are in same phase, these two fields are perpendicular to the direction of propagation, hence transverse in nature.
2. Changes in electric and magnetic fields occur at the same time and place, hence are in phase.
3. EM waves do not require any material medium for their propagation; it can pass through vacuum with a speed $c = 3 \times 10^8 \text{ ms}^{-1}$.
4. Incident electromagnetic wave exhibits phenomena like reflection, refraction, diffraction, interference and polarization.

I. **Use of Radio waves are :**

1. For Television program transmission
2. For Radio Program transmission (Television uses higher frequencies than radio)

II. **Use of Microwaves are :**

1. For cooking purpose in microwave ovens
2. Satellite television programs, mobile phones, in radar systems

Q24. In a plot of photoelectric current versus anode potential, how does

- (I) The saturation current vary with anode potential for incident radiations of different frequencies but same intensity?
- (II) The stopping potential vary for incident radiations of different intensities but same frequency?
- (III) Photoelectric current vary for different intensities but same frequency of incident radiations? Justify your answer in each case.

Answer:

- (I) When intensity remains same, saturation current is independent of frequencies. It does not change with the increase in anode potential.
- (II) If frequency remains same, stopping potential is independent of intensities. Since stopping potential depends on frequency and not on the intensity of incident radiation.
- (III) Photoelectric current increases linearly with the intensity of incident radiation of same frequency.



Q25. Calculate the amount of energy released during the α – decay of ${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + {}^4_2\text{He}$

[Given : 1. atomic mass of ${}^{238}_{92}\text{U} = 238.05079 \text{ u}$

2. atomic mass of ${}^{234}_{90}\text{Th} = 234.04363 \text{ u}$

3. atomic mass of ${}^4_2\text{He} = 4.00260 \text{ u}$

$1 \text{ u} = 931.5 \text{ MeV}/c^2$]

Is this decay spontaneous? Give reason.

Answer: Difference in mass will be converted to energy, 1 amu gives 931.5 MeV energy.

The energy released in the α – decay is

$$\begin{aligned}\Delta E &= [m({}^{238}_{92}\text{U}) - m({}^{234}_{90}\text{Th}) - m({}^4_2\text{He})]c^2 \\ &= [238.05079 - 234.04363 - 4.00260] \times 931.5 \text{ MeV} \\ &= 0.00456 \times 931.5 = 4.25 \text{ MeV}\end{aligned}$$

Spontaneous decay depends on ΔE value, since ΔE is positive, the decay process is spontaneous.

Q26. What is a digital signal? Write two advantages of digital communication. Give any one difference between Fax and E-mail systems of communication.

Answer: A digital signal refers to an electrical signal that is converted into a pattern of bits. Unlike an analog signal, which is a continuous signal that contains time-varying quantities, a digital signal has a discrete value at each sampling point. Hence current or voltage take only two discrete values 0 and 1 in digital signal.

Advantages of digital communication:

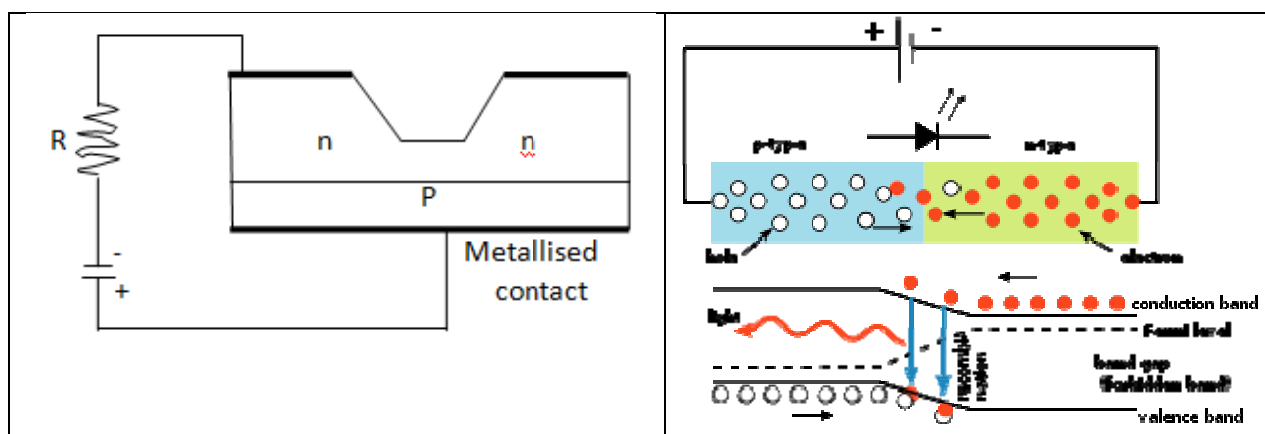
- (I) Digital signals can convey information with greater noise immunity, because each information component (byte etc) is determined by the presence or absence of a data bit (0 or one) This mode of communication is more reliable.
- (II) Enables transmission of signals over a long distance and Its transmission needs simple technique.

We get a hard copy as printout at recipient's terminal in fax communication where as we get a soft copy at the recipient's terminal in email communication.



Q27. Explain, with the help of a schematic diagram, the principle and working of a Light Emitting Diode. What criterion is kept in mind while choosing the semiconductor material for such a device? Write any two advantages of Light Emitting Diode over conventional incandescent lamps.

Answer: A light-emitting diode (LED) is a two-lead semiconductor light source. It is a basic pn-junction (forward biased) diode, which emits light when activated. When a suitable voltage is applied to the leads, electrons are able to recombine with electron holes within the device, releasing energy in the form of photons. This effect is called electroluminescence, and the color of the light (corresponding to the energy of the photon) is determined by the energy band gap of the semiconductor.



In a forward biased p-n junction, the electrons of the n-region and holes of p-region are pushed towards the junction where electron-hole recombination takes place. As the electrons are in higher conduction band on n-side and holes are in lower valence band on p-side, the energy difference appears in the form of heat or light radiation during the process of recombination.

Based on the requirement of wavelength of light required to be emitted from LED suitable semiconductor is chosen.

Advantages:

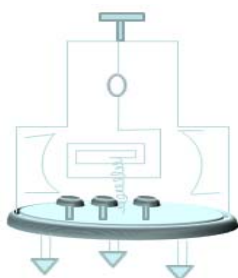
- (I) LEDs emit more lumens per watt than incandescent light bulbs. Efficiency of LED's are high in comparison to bulbs or tubes..
- (II) It takes micro seconds to warm up and lifetime of LED is very high 35,000 to 50,000 hrs.



Q28. Draw a labeled diagram of a moving coil galvanometer. State the principle on which it works.

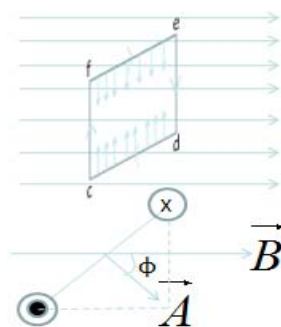
Deduce an expression for the torque acting on a rectangular current carrying loop kept in a uniform magnetic field. Write two factors on which the current sensitivity of a moving coil galvanometer depends.

Answer:



Principle : (Based on the principle of the force of interaction between the current carrying conductor and a magnetic field) A coil is suspended in a magnetic field. The current to be measured is passed through the coil. The current through the coil produces a magnetic field which interacts with the given magnetic field in which the coil is suspended. Due to the interaction the coil gets deflected and measuring the angle of rotation current can be calculated.

A circular or rectangular coil of about 10 to 15 turns of a fine insulated copper or aluminium wire is suspended from a torsion head T, by means of quartz fibre in between the concave pole pieces of a strong magnet. The lower end of the coil is attached to a light springs which brings the coil back to its original position when the current is stopped. The suspension wire and the spring are connected with two terminal screws at the base which acts as the leads of current. The whole thing is enclosed in a metal box provided with glass face and is supported on leveling screws. The angle of rotation of the coil is measured by lamp & scale arrangement, reflection takes place from the tiny mirror M attached with the suspension wire.



Theory : Given n = the number of turns in the coil

l & b = the length of each vertical & horizontal side of the coil respectively (assuming) the coil to be rectangular
 i = the current flowing through the coil.

B = the induction vector of the magnetic field in which the coil is suspended.

We know that the force experienced by the current carrying conductor placed in a magnetic field is

$$\vec{F} = i(\vec{l} \times \vec{B}) \rightarrow (1)$$

Force on each wire 'de' and 'fc'

$$|\vec{F}| = ibB \sin(90 - \phi) = ibB \sin \phi$$

where ϕ = angle between the normal to the plane of the coil and the direction of the magnetic field.

The direction of the force on the wires 'de' & 'fc' are in the plane of the coil along the downward and upward direction respectively and hence cancel out.

The force on the two vertical wires 'cd' and 'ef' by using equation(1) are found to have magnitude

$$F = i/B \sin 90^\circ = i/b$$



Applying the right hand curl rule for vector product the directions of the force are as shown in the figure. These two forces being equal in magnitude, opposite in direction parallel and non co-planer constitute a couple.

Moment of the couple $\Gamma = iB \times OC$

$$\Gamma = iBb \sin \phi = i(lb)B \sin \phi = iAB \sin \phi$$

Since there are n turns in the coil the total torque experienced by the current carrying coil in the magnetic field

$$\Gamma = niAB \sin \phi$$

Due to this torque the coil rotates a restoring couple due to the torsion rigidity sets in and when the restoring couple equals to the deflecting couple the coil comes to equilibrium



Let c = torsional couple per unit twist of the suspension wire

θ = the angle through which the coil rotate in the position of equilibrium

$$\text{Restoring couple} = c\theta \rightarrow (1)$$

$$\text{From equation(2) and (3): } nAiB \sin \phi = c\theta$$

$$i = \frac{c}{nAB} \frac{\theta}{\sin \phi} \rightarrow (4)$$

By using concave pole pieces the magnetic field is made

radial so that for any position of coil ϕ is always 90° ,

$$\sin \phi = \sin 90^\circ = 1$$

$$\therefore i = \frac{c}{nAB} \theta \rightarrow (5)$$

Since c, n, A & B are all constant

$$i \propto \theta$$

The current sensitivity of a moving coil galvanometer depends on

- (I) Number of turns in the galvanometer coil.
- (II) Torsional constant of the suspension fibre.



Or,

State Biot-Savart law. Use it to derive an expression for the magnetic field at the center of a circular loop of radius R carrying a steady current I . Sketch the magnetic field lines for such a current carrying loop.

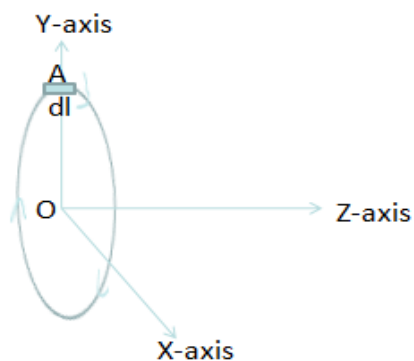
Answer: **Biot-Savart Law** : It states that the magnetic field $d\vec{B}$ due to a current element $d\vec{l}$ carrying a steady current I at a point P at a distance r from the current element is

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3}$$

$$\Rightarrow dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$$

Where θ is the angle between the direction of the current and the line joining the current element.

Magnetic Field at the center of the circular Coil/Loop Carrying Current:



We have a circular coil of many turns a current flows through the coil. We have to find the magnetic field at the centre of the coil due to the current through the coil.

Given

i = current flowing through the coil

a = radius of the coil

n = number of turns in the coil

Let us choose a right handed system of mutually perpendicular axes X, Y and Z as follows. The plane of the coil is represented by $Y-Z$ plane and X -axis is perpendicular to the plane of the coil at O .

Let us consider an element of the coil at A of length dl . The axis of the element is along Z -axis i.e. dl is along Z -axis using Biot's law the magnetic field at O due to the current element at A

$$d\vec{b} = \frac{\mu_0}{4\pi} \frac{i}{r^2} (\vec{dl} \times \hat{r}) \rightarrow (1)$$

Where $r = a$ = radius of the coil

$\hat{r}$ = a unit vector along Y axis

Taking the magnitude of equation(1)

$$|d\vec{b}| = db = \frac{\mu_0}{4\pi} \frac{i}{a^2} dl \times 1 \times \sin 90^\circ = \frac{\mu_0}{4\pi} \frac{idl}{a^2} \rightarrow (2)$$



Applying the right handed curl rule for vector product the direction of the magnetic field is along X axis in the figure. If we consider the element at different points along the coil it is found that for every position of the element the direction of the magnetic field at the centre O along X-axis. Since the direction of the magnetic field at O is same for every position of the element hence vector addition reduces to scalar addition and resultant magnetic field at O can be obtained by integrating equation(2)

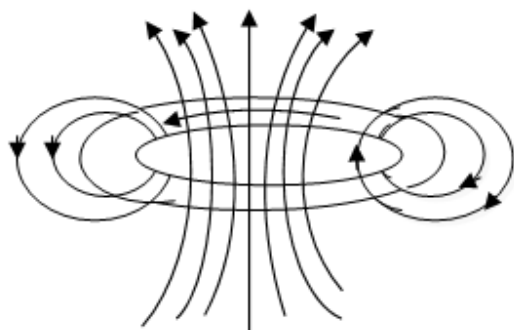
$$B = \int dB = \int \frac{\mu_0}{4\pi} \frac{idl}{a^2} = \frac{\mu_0}{4\pi} \frac{i}{a^2} \int dl = \frac{\mu_0}{4\pi} \frac{i}{a^2} (2\pi a \times n) = \frac{\mu_0}{2} \frac{ni}{a} \rightarrow (3)$$

$$\int dl = \text{the total length of the coil} = (2\pi a)n$$

Equation (3) gives the magnitude of the magnetic field at the centre of the circular coil carrying current.

Direction of Magnetic field:

The direction of the magnetic field at the centre which is found to be along X axis can be stated as follows :



"Curl the finger of right hand in the direction of current in the coil the thumb gives the direction of the magnetic field"

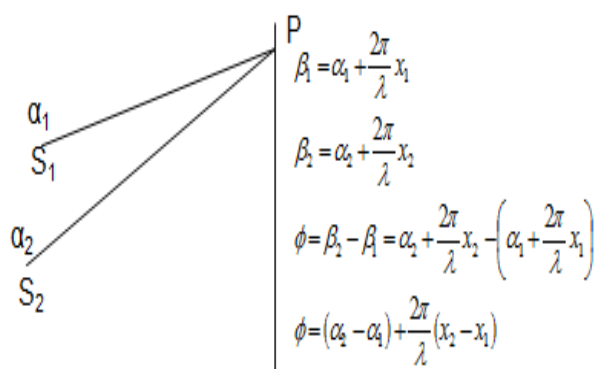


Q29. What are coherent sources? Why are coherent sources required to produce interference of light? Give an example of interference of light in everyday life.

In Young's double slit experiment, the two slits are 0.03 cm apart and the screen is placed at a distance of 1.5 m away from the slits. The distance between the central bright fringe and fourth bright fringe is 1 cm. Calculate the wavelength of light used.

Answer: Two sources of light which continuously emit light waves of same frequency with a zero or constant phase difference between them are called **coherent sources**. Two independent sources such as two bulbs or two candles or two lamps can never be coherent source. Coherent sources are always produced from a single source in two different ways.

1. By division of wave front
2. By division of amplitude



$$\phi = (\alpha_2 - \alpha_1) + \frac{2\pi}{\lambda} (x_2 - x_1)$$

$$I = \text{constant} [a^2 + b^2 + 2ab \cos \phi]$$

If the sources are not coherent the initial phase difference ($\alpha_2 - \alpha_1$) changes with time 10^8 times per second, i.e. intensity (I) at every point changes 10^8 times per second. Since human eye can not follow such quick change hence intensity at every point appears to be same i.e. no fringes can be seen. But if the sources are coherent initial phase are same $\alpha_2 - \alpha_1 = 0$. Hence Coherent sources are necessary to produce sustained interference pattern.

Example: Oil spread over water (thin film) shows beautiful colours due to interference of light.

Numerical Problem: Distance of n th bright fringe from the central bright fringe is $x_n = \frac{nD\lambda}{d}$

Hence distance of fourth bright fringe ($n=4$) from the central bright fringe is

$$x_4 = \frac{4D\lambda}{d}$$

$$\therefore \lambda = \frac{dx_4}{4D} = \frac{0.03 \times 10^{-2} \times 1 \times 10^{-2}}{4 \times 1.5}$$

$$= 5 \times 10^{-7} \text{ m}$$



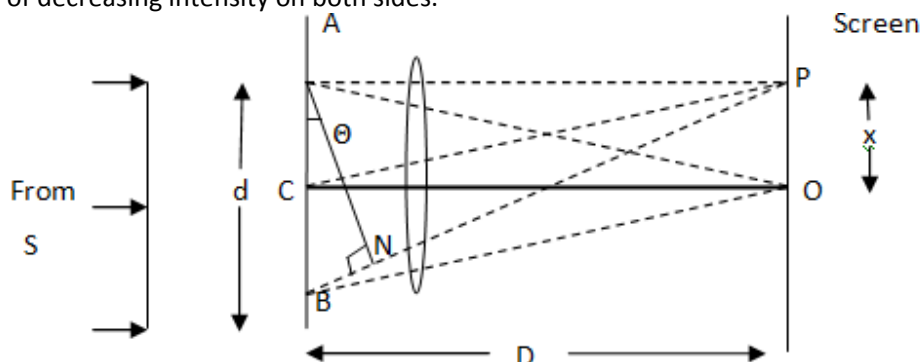
Or, State the condition under which the phenomenon of diffraction of light takes place. Derive an expression for the width of the central maximum due to diffraction of light at a single slit.

A slit of width 'a' is illuminated by a monochromatic light of wavelength 700 nm at normal incidence. Calculate the value of 'a' for position of

- (I) First minimum at an angle of diffraction of 30° .
- (II) First maximum at an angle of diffraction of 30° .

Answer: Condition for Diffraction (bending of light) to take place, is light in its path should face small obstacle which are either (i) round a sharp edge (ii) a narrow opening. Sharp narrow or small means sizes should be comparable to the wave length of light. Thus Diffraction of light takes place when the width of the slit is comparable to the wavelength of light used.

Expression for width of central maximum: Considering a parallel beam of monochromatic light of wavelength λ falls normally on a slit AB of width d (of the order of the wave length of light). While passing through the slit diffraction occurs. The diffraction pattern is focused on to the screen by a convex lens. The diffraction pattern consists of a central bright fringe (or band), having alternate dark and bright fringes of decreasing intensity on both sides.



Position of central maximum: Let C be the center of the slit AB. According to Huygen's principle, when light falls on the slit, it becomes a source of secondary wavelets. All the wavelets originating from slit AB are in same phase. These secondary waves reinforce each other resulting the central maximum intensity at O.

- 1. Position of secondary Maxima and Minima:** Considering a point P on the screen. All the secondary waves travelling in a direction making angle θ with CO, reach at a point P. The intensity at P depends on the path difference between secondary waves. Therefore, path difference between the secondary waves reaching P from points A and B is

$$BN = d \sin \theta$$

- (I) The point P will be the position of **nth secondary maxima** if path difference

$$BN = (2n + 1) \frac{\lambda}{2}, \text{ where } n \text{ is an integer.}$$



$$\therefore d \sin \theta = (2n + 1) \frac{\lambda}{2}$$

$$\sin \theta = (2n + 1) \frac{\lambda}{2d}$$

Where θ is very small then $\sin \theta = \theta$

$$\therefore \theta = (2n + 1) \frac{\lambda}{2d}, \text{ when } n \text{ is an integer.}$$

for $n = 1, \theta_1 = \frac{3\lambda}{2d}$, first secondary maximum.

(II) The point P will be the position of **nth secondary minima** if path difference

$$BN = n\lambda$$

$d \sin \theta = n\lambda$, where n is an integer.

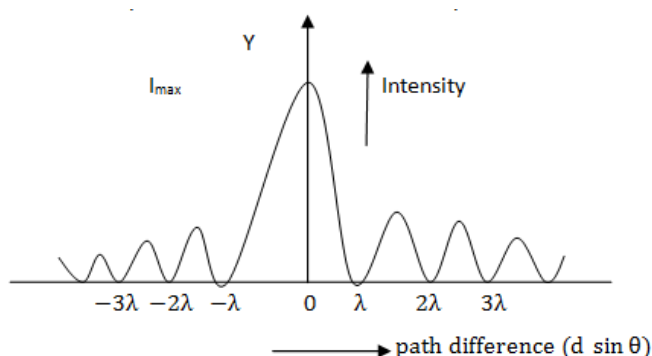
$$\sin \theta = \frac{n\lambda}{d}$$

Where θ is very small then $\sin \theta = \theta$

$$\therefore \theta = \frac{n\lambda}{d}, \text{ when } n \text{ is an integer.}$$

Hence the diffraction pattern due to single slit consists of a center bright maximum at P along with secondary maxima and minima on either side.

Intensity Distribution: The intensity distribution on the screen is represented as



Width of central maximum: If d is the distance between slit and screen and y_n is the distance of n th minimum from point O then

$$\theta_n = \frac{y_n}{D} \dots \dots \dots (i)$$

But the position of n th secondary minimum is

$$\theta_n = \frac{n\lambda}{D} \dots \dots \dots (ii)$$

$$\therefore \frac{y_n}{D} = \frac{n\lambda}{D} \quad [\text{From (i) \& (ii)}]$$

$$\beta = y_n - y_{n-1} = \frac{n\lambda D}{d} - (n-1) \frac{\lambda D}{d} = \frac{n\lambda D - n\lambda D + \lambda D}{d}$$

$$\therefore \beta = \frac{\lambda D}{d}$$



Problem: (I) Condition for first minimum is

$$a \sin \theta = \lambda$$

$$a = \frac{\lambda}{\sin \theta} = \frac{700 \text{ nm}}{\sin 30^\circ} = \frac{700 \text{ nm}}{0.5} = 1400 \text{ nm}$$

(II) Condition for first maximum is

$$a \sin \theta = \frac{3\lambda}{2}$$

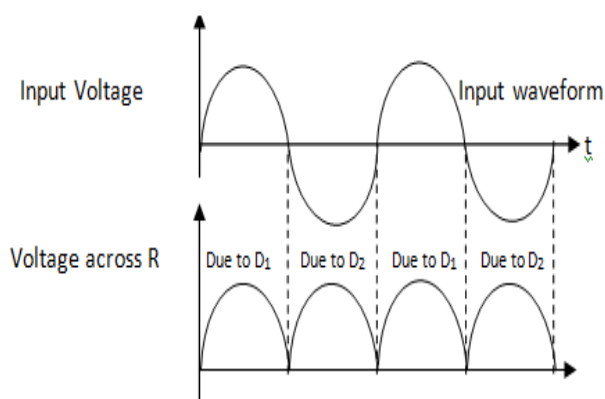
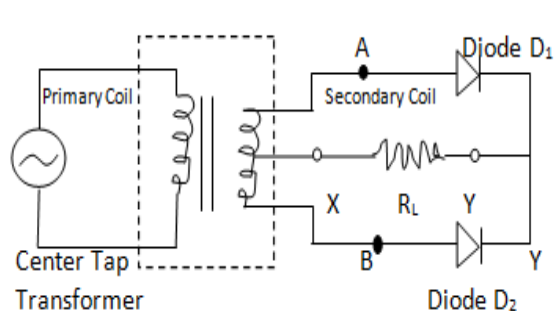
$$\therefore a = \frac{3\lambda}{2 \sin \theta} = \frac{3 \times 700 \text{ nm}}{2 \times \sin 30^\circ} = 2100 \text{ nm}$$

Q30. State the principle of working of p-n diode as a rectifier. Explain, with the help of a circuit diagram, the use of p-n diode as a full wave rectifier. Draw a sketch of the input and output waveforms.

Answer: p-n junction diode as full wave rectifier: For full wave rectifier two diodes are used so that during one half cycle of A.C. one diode (say D_1) is operational during the other half cycle second diode (D_2) is operational. A center tap transformer is used as shown in the circuit. The secondary transformer gives the desired A.C. voltage across A and B.

During the positive half cycle of A.C. input, the diode D_1 is in forward bias and conduct current while D_2 is in reverse biased and does not conduct current. So we get an output voltage across the load resistor R_L .

During negative half cycle of A.C. input, the diode D_1 is in reverse biased and does not conduct current while diode D_2 is in forward biased and conducts current. So we get an output voltage across the load resistor R_L .



$I = I_0 \sin \omega t$ be the input current to be rectified.

$$\therefore \text{The average current} = \frac{2I_0}{\pi}$$

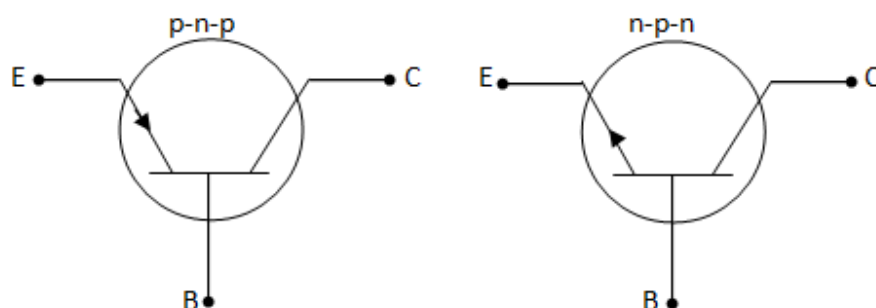
$$\text{and the output voltage} = \frac{2I_0}{\pi} R_L$$



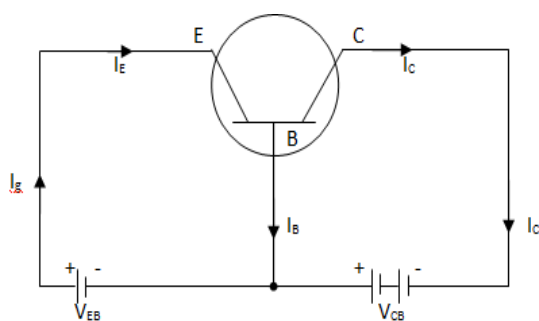
Or,

Draw the symbolic representation of a (I) p-n-p, (II) n-p-n transistor. Why is the base region of transistor thin and lightly doped? With proper circuit diagram, show the biasing of a p-n-p transistor in common base configuration. Explain the movement of charge carriers through different parts of the transistor in such a configuration and show that $I_E = I_C + I_B$.

Answer: The symbols of p-n-p and n-p-n transistors



Thin and lightly doped base region contains very few majority charge carriers. This reduces the recombination rate of electrons and holes across the emitter-base junction. Also increases collector current and hence increases current gain of the transistor.



Action of p-n-p transistor: The emitter base junction is forward biased by battery V_{EB} and collector-base junction is reversed biased by battery V_{CB} . The positive terminal of the battery V_{EB} repels the holes of the emitter towards the base about 5% of the holes recombine with the electrons of the base. The remaining 95%(approx) of the holes enter the collector region under the reverse bias. As the holes reach the end of the collector region, they attract the electrons from the negative terminal of the battery V_{CB} and combine with them. At the same time covalent bonds are broken in the emitter region, and an equal number of electrons get attracted towards the positive terminal of V_{EB} and holes flow towards the base. Hence Emitter current is sum of base and collector current i.e. $I_E = I_B + I_C$.